

Shenggan Cheng

+(86) 13636684726+(65) 98854484

shenggan@comp.nus.edu.sg shenggan.c@outlook.com

EDUCATION

Ph.D. in Computer Science 2022/01 – 2026/07

National University of Singapore, Singapore

Advisor: Yang You

B.E. in Computer Science 2015/09 – 2019/06

Shanghai Jiao Tong University, China

Advisor: James Lin

INDUSTRY EXPERIENCE

HUAWEI Zurich Research Center, Zurich, Switzerland 2026/07 – now

Researcher, Computing Systems Lab

Microsoft Research Asia, Remote 2025/06 – 2025/12

Research Intern, Machine Learning Group

Alibaba Cloud, Beijing, China 2023/04 – 2025/06

Research Intern for Alibaba Innovative Research, PAI Research team

- *Automated Communication Optimization and Scheduling*
- *Diffusion Inference Acceleration*

Center for HPC | Shanghai Jiao Tong University, Shanghai, China 2019/07 – 2021/02

Research Engineer

- *Parallel Cosmology N-Body Simulation for HPC Cluster*

SenseTime, Shanghai, China 2017/12 – 2019/03

Research Intern, SenseAuto Team

- *Prediction of Pedestrians and Vehicle Trajectory*

PUBLICATION

1. **Shenggan Cheng***, Shengjie Lin*, Lansong Diao, Hao Wu, Siyu Wang, Chang Si, Ziming Liu, Xuanlei Zhao, Jiangsu Du, Wei Lin, Yang You. *Concerto: Automatic Communication Optimization and Scheduling for Large-*

Scale Deep Learning. *Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2025.

2. **Shenggan Cheng**, Xuanlei Zhao, Guangyang Lu, Jiarui Fang, Tian Zheng, Ruidong Wu, Xiwen Zhang, Jian Peng, Yang You. **FastFold: Optimizing AlphaFold Training and Inference on GPU Clusters**. *Principles and Practice of Parallel Programming (PPoPP)*, 2024.
3. Ziming Liu*, **Shenggan Cheng***, Haotian Zhou, Yang You. **Hanayo: Harnessing Wave-like Pipeline Parallelism for Enhanced Large Model Training Efficiency**. *The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, 2023.
4. **Shenggan Cheng***, Hao-Ran Yu*, Derek Inman, Qiucheng Liao, Qiaoya Wu, James Lin. **CUBE--Towards an Optimal Scaling of Cosmological N-body Simulations**. *International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, 2020.
5. Geng Zhang, **Shenggan Cheng**, Xuanlei Zhao, Ziming Liu, Yang You. **HelixPipe: Efficient Distributed Training of Long Sequence Transformers with Attention Parallel Pipeline Parallelism**. *Principles and Practice of Parallel Programming (PPoPP)*, 2026.
6. Zedong Liu, **Shenggan Cheng**, Guangming Tan, Yang You, Dingwen Tao. **ElasticMM: Efficient Multimodal LLMs Serving with Elastic Multimodal Parallelism**. *Annual Conference on Neural Information Processing Systems (NeurIPS Oral)*, 2025.
7. Xuanlei Zhao, **Shenggan Cheng**, Chang Chen, Zangwei Zheng, Ziming Liu, Zheming Yang, Yang You. **DSP: Dynamic Sequence Parallelism for Multi-Dimensional Transformers**. *International Conference on Machine Learning (ICML)*, 2025.
8. Xuanlei Zhao, **Shenggan Cheng**, Guangyang Lu, Haotian Zhou, Bin Jia, Yang You. **AutoChunk: Automated Activation Chunk for Memory-Efficient Long Sequence Inference**. *International Conference on Learning Representations (ICLR)*, 2024.
9. Binrui Li, **Shenggan Cheng**, and James Lin. **tcFFT: Accelerating Half-Precision FFT through Tensor Cores**. *IEEE International Conference on Cluster Computing (CLUSTER)*, 2021.
10. Ziming Liu, Shaoyu Wang, **Shenggan Cheng**, Zhongkai Zhao, Kai Wang, Xuanlei Zhao, James Demmel, Yang You. **StarTrail: Concentric Ring Parallelism for Efficient Near-Infinite-Context Transformer Model Training**. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
11. Yong Liu, Di Fu, **Shenggan Cheng**, Zirui Zhu, Yang Luo, Minhao Cheng, Cho-Jui Hsieh, Yang You. **SeedLoRA: A Fusion Approach to Efficient LLM Fine-Tuning**. *International Conference on Machine Learning (ICML)*, 2025.
12. Jinhui Wei, **Shenggan Cheng**, Wei Zhu, Jiazhi Jiang, Dan Huang, Zhiguang Chen, Jiangsu Du, Yutong Lu. **Dynamic Latency-Throughput Balancing in Distributed Large Model Inference with Interleaved Parallelism**. *ACM Transactions on Architecture and Code Optimization (TACO)*, 2026.
13. Jiangsu Du, Jinhui Wei, Jiazhi Jiang, **Shenggan Cheng**, Dan Huang, Zhiguang Chen, Yutong Lu. **Liger: Interleaving Intra- and Inter-Operator Parallelism for Distributed Large Model Inference**. *Principles and Practice of Parallel Programming (PPoPP)*, 2024.
14. Xuanlei Zhao, Bin Jia, Haotian Zhou, Ziming Liu, **Shenggan Cheng**, Yang You. **HeteGen: Efficient Heterogeneous Parallel Inference for Large Language Models on Resource-Constrained Devices**. *Conference on Machine Learning and Systems (MLSys)*, 2024.

15. Jinxiao Zhang, Xiyong Wu, Runmin Dong, Xihan Huang, **Shenggan Cheng**, Yunkai Yang, Zheng Zhou, Yunpu Xu, Zhaoyang Luo, Fan Wei, Miao Yang, Mengxuan Chen, Fan Yang, Bin Luo, Conghui He, Haohuan Fu. [Scaling Spatiotemporal Generative Compression for Earth Observation on an HPC-AI-Converged Supercomputer](#). *The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, 2026.
16. Ziming Liu, Boyu Tian, Guoteng Wang, Zhen Jiang, Peng Sun, Zhenhua Han, Tian Tang, Xiaohe Hu, Yanmin Jia, Yan Zhang, He Liu, Mingjun Zhang, Yiqi Zhang, Qiaoling Chen, **Shenggan Cheng**, Mingyu Gao, Yang You, Siyuan Feng. [Expert-as-a-Service: Towards Efficient, Scalable, and Robust Large-scale MoE Serving](#). *The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, 2026.
17. Yuanxin Wei, Lansong Diao, Bujiao Chen, **Shenggan Cheng**, Zhengping Qian, Wenyan Yu, Nong Xiao, Wei Lin, Jiangsu Du. [MixCache: Mixture-of-Cache for Video Diffusion Transformer Acceleration](#). *ACM International Conference on Multimedia Retrieval (ICMR)*, 2026.
18. Kunyuan Du, Ya Zhang, Haibing Guan, Qi Tian, **Shenggan Cheng**, James Lin. [FTL: A universal framework for training low-bit DNNs via Feature Transfer](#). *European Conference on Computer Vision (ECCV)*, 2020.
19. Zixuan Huang, Junming Fan, **Shenggan Cheng**, Shuai Yi, Xiaogang Wang, Hongsheng Li. [HMS-Net: Hierarchical Multi-Scale Sparsity-Invariant Network for Sparse Depth Completion](#). *IEEE Transactions on Image Processing (TIP)*, 2019.
20. **Shenggan Cheng**, Yuanxin Wei, Lansong Diao, Yong Liu, Bujiao Chen, Lianghua Huang, Yu Liu, Wenyan Yu, Jiangsu Du, Wei Lin, Yang You. [SRDiffusion: Accelerate Video Diffusion Inference via Sketching-Rendering Cooperation](#). *arXiv*, 2025.
21. Jinxiao Zhang, Yunpu Xu, Xiyong Wu, Runmin Dong, **Shenggan Cheng**, Yi Zhao, Mengxuan Chen, Qinrui Zheng, Jianting Liu, Haohuan Fu. [DiT-HC: Enabling Efficient Training of Visual Generation Model DiT on HPC-oriented CPU Cluster](#). *arXiv*, 2026.
22. Jinxiao Zhang, Runmin Dong, Xiyong Wu, Xihan Huang, **Shenggan Cheng**, Yunkai Yang, Zheng Zhou, Yunpu Xu, Zhaoyang Luo, Miao Yang, Fan Wei, Mengxuan Chen, Yang You, Juepeng Zheng, Weijia Li, Yutong Lu, Haohuan Fu. [Transforming the Use of Earth Observation Data: Exascale Training of a Generative Compression Model with Historical Priors for up to 10,000 x Data Reduction](#). *arXiv*, 2026.
23. **Shenggan Cheng**, Ziming Liu, Jiangsu Du, Yang You. [ATP: Adaptive Tensor Parallelism for Foundation Models](#). *arXiv*, 2023.

SERVICE

The Forty-First AAAI Conference on Artificial Intelligence(AAAI), 2027	PC Member
Supercomputing Asia(SCA), 2027	Poster PC Member
Supercomputing Asia (SCA) / International Conference on High Performance Computing in Asia-Pacific Region (HPCAsia), 2026	PC Member
International Workshop on ARM-Based System for Science @ SCA, 2024	Local Chair
Annual Conference on Neural Information Processing Systems (NeurIPS), 2026	Reviewer
Conference on Computer Vision and Pattern Recognition (CVPR), 2026	Reviewer
International Conference on Learning Representations (ICLR), 2026	Reviewer

European Conference on Computer Vision (ECCV), 2026	Reviewer
International Conference on Parallel Processing (ICPP), 2024	Reviewer
Journal of Computer Science and Technology	Reviewer

AWARDS

ASC18 Student Supercomputer Competition	First Prize (4th Worldwide) Application Innovation Award	2018/03
The 5th Student RDMA Programming Competition	First Prize	2017/10
The Interdisciplinary Contest in Modeling	Meritorious Winner	2017/02

TEACHING & MENTORSHIP

Advanced Topics in Machine Learning National University of Singapore, PhD-level, CS6216	Teaching Assistant	2024
Neural Networks and Deep Learning National University of Singapore, graduate-level, CS5242	Teaching Assistant	2023
ASC Student Supercomputer Challenge Shanghai Jiao Tong University, Global Championship	Consulting	2025
8th Intel Cup Parallel Application Challenge Shanghai Jiao Tong University, Excellent Paper Award & Bronze Award (Application Track)	Advisor	2020